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**PROGRAM RISET KOLABORASI INDONESIA-EQUITY**  
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**Pengembangan dan Validasi Model Pembelajaran VALORIZE:  
Kerangka Pedagogis Berbasis AI untuk Membentuk Identitas  
Profesional Rekayasawan di Era Digital**

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## RINGKASAN EKSEKUTIF

Profesi rekayasawan (insinyur) saat ini menghadapi krisis eksistensial akibat otomatisasi dari Kecerdasan Buatan (AI), di mana diperkirakan 47% dari tugas teknik rutin akan diotomatisasi dalam lima tahun ke depan. Kondisi ini mengungkap kesenjangan pedagogis inti dalam pendidikan tinggi, yakni fokus yang berlebihan pada penguasaan konten (**know-ing**) sambil mengabaikan kompetensi khas manusia, seperti penalaran etis dan pemecahan masalah kompleks (**becoming**). Riset ini bertujuan mengembangkan dan memvalidasi kerangka kerja pedagogis yang tangguh terhadap otomasi melalui sistem **Collaborative Knowledge Management System-Smart Engineering** (CKMS-SE) sebagai perwujudan operasional prinsip TISE (**Triune-Intelligence Smart Engineering**) dan **Valorize**. Dalam paruh waktu pelaksanaan, riset telah merumuskan landasan teori berdasarkan **Cognitive Load Theory** (CLT) dan **Dual Coding Theory** (DCT), serta berhasil melakukan uji kuasi-eksperimental pada 142 mahasiswa. Hasil menunjukkan lonjakan signifikan pada motivasi intrinsik (Cohen's  $d = 0.97$ ), identitas profesional ( $d = 0.84$ ), dan pembelajaran transfer ( $d = 0.95$ ). Platform ini juga mendokumentasikan penciptaan 427 artefak berharga dengan 16% tingkat penggunaan ulang oleh kohort berikutnya. Capaian luaran sejauh ini meliputi sebuah draf artikel jurnal internasional yang kini berada pada status **Under Review** di penerbit Frontiers (Jurnal Q2 Scopus).

# BAB 1 PENDAHULUAN

## 1.1 LATAR BELAKANG MASALAH

Pendidikan teknik selama ini bertumpu pada pendekatan berbasis konten, di mana mahasiswa menghabiskan sebagian besar waktunya mempelajari rumus matematika, algoritma, dan desain perangkat lunak rutin. Namun, Laporan Forum Ekonomi Dunia 2023 memperkirakan bahwa 47% tugas teknik saat ini akan menghadapi otomatisasi dari mesin dalam waktu lima tahun ke depan.

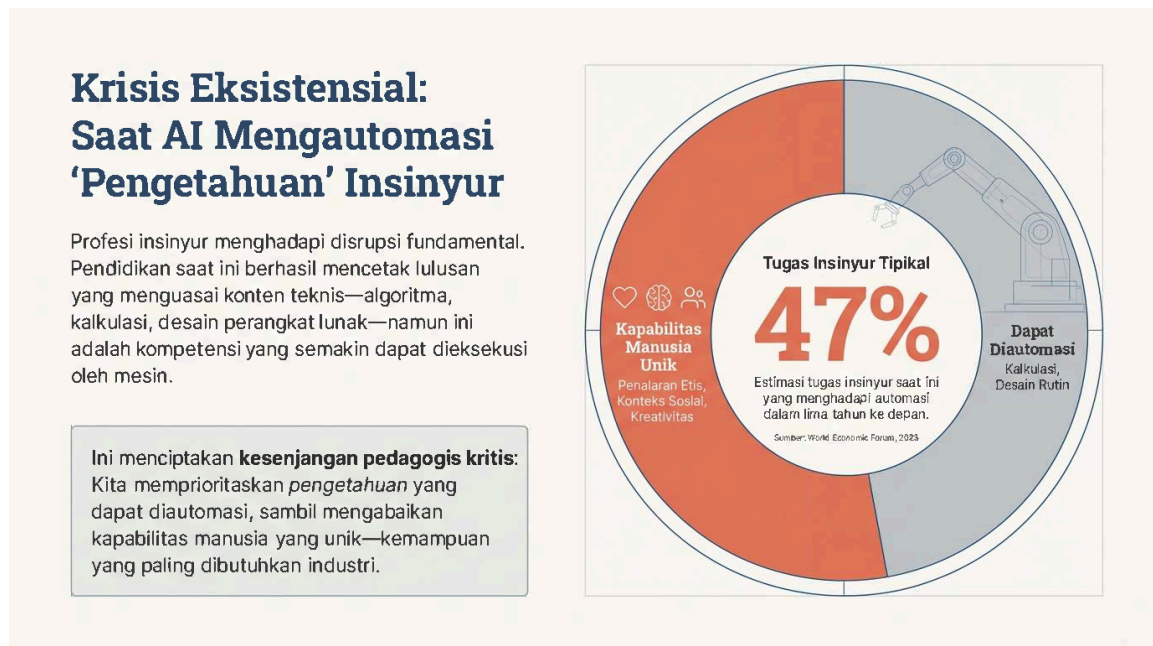


Figure 1: Krisis Eksistensial Saat AI Mengotomasi 'Pengetahuan' Insinyur

Otomatisasi ini mengungkap kesenjangan kritis: pendidikan kita terlalu memprioritaskan “mengetahui” (*knowing*), sebuah kemampuan yang kian mudah direplikasi oleh AI, namun mengabaikan pembentukan “menjadi” (*becoming*), yakni kemampuan yang berakar pada nilai-nilai manusia yang unik seperti etika, kolaborasi, dan empati (prinsip *Homocordium*). Lebih jauh, dari perspektif psikologi kognitif, pendekatan tradisional menimbulkan beban kognitif yang masif. Domain sistem dan sinyal yang sarat konsep dan matematika sering kali membebani memori kerja mahasiswa, yang secara alamiah hanya mampu menyimpan 3 hingga 7 potong informasi dalam waktu 15 hingga 30 detik. Jika lingkungan pembelajaran hanya menambah *extraneous load* (beban kognitif eksternal yang tidak relevan), proses pembentukan skema mental atau keahlian sejati akan sangat terhambat.

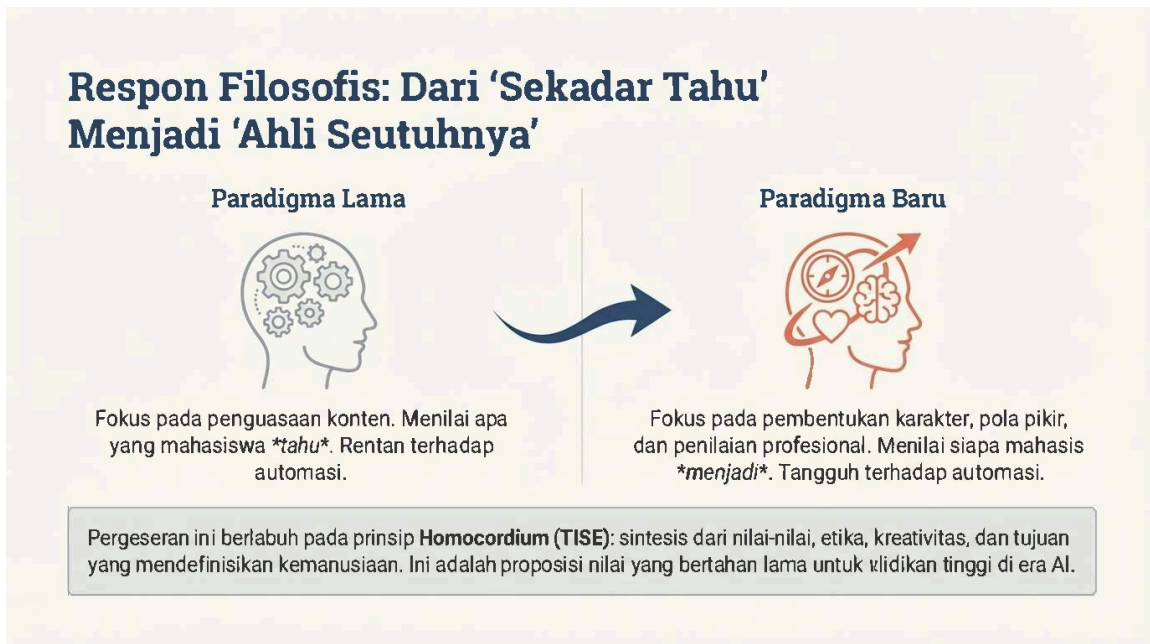


Figure 2: Respon Filosofis: Dari 'Sekadar Tahu' Menjadi 'Ahli Seutuhnya'

Terdapat perbandingan antara Paradigma Lama dan Paradigma Baru.

- **Paradigma Lama (Knowing / Sekadar Tahu):** Fokus pada penguasaan konten. Sistem ini menilai apa yang mahasiswa *tahu*, yang mana keahlian ini sangat rentan terhadap automasi.
- **Paradigma Baru (Becoming / Ahli Seutuhnya):** Fokus pada pembentukan karakter, pola pikir, dan penilaian profesional. Sistem ini menilai siapa mahasiswa tersebut akan *menjadi*. Karakteristik ini tangguh dan tahan terhadap ancaman automasi. Pergeseran fundamental ini berlabuh pada prinsip **Homocordium (TISE)**: yaitu sebuah sintesis dari nilai-nilai, etika, kreativitas, dan tujuan yang mendefinisikan kemanusiaan. Ini adalah proposisi nilai yang bertahan lama untuk pendidikan tinggi di era Kecerdasan Buatan (AI).

## Memperkenalkan CKMS-SE: Cetak Biru untuk Filosofi yang Dapat Dieksekusi

Untuk menjawab tantangan ini, kami mengembangkan **CKMS-SE**: sebuah *Collaborative Knowledge Management System for Smart Engineering*. Sistem ini bukan sekadar alat, melainkan validasi empiris dan perwujudan operasional pertama dari filosofi TISE-Valorize.

Apakah sistem manajemen pengetahuan kolaboratif yang dirancang melalui prinsip TISE-Valorize, yang menggabungkan pemodelan kognitif formal dan alur kerja profesional, dapat meningkatkan pembentukan identitas profesional dan penciptaan nilai yang terukur dalam pendidikan teknik?

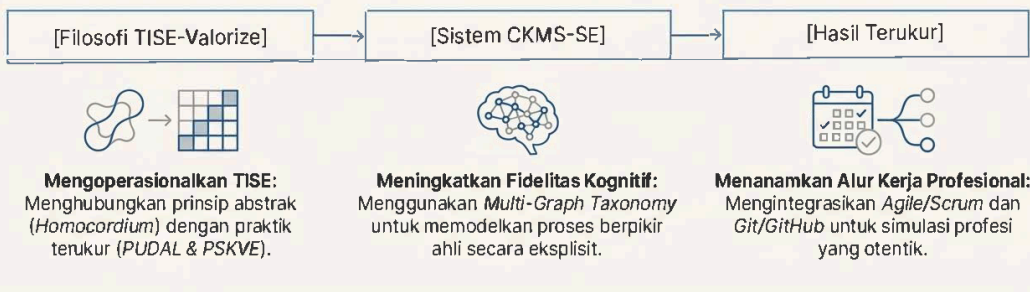


Figure 3: Memperkenalkan CKMS-SE: Cetak Biru untuk Filosofi yang Dapat Dieksekusi

Untuk menjawab tantangan tersebut, riset ini mengembangkan **CKMS-SE** (*Collaborative Knowledge Management System for Smart Engineering*). Sistem ini bukan sekadar alat, melainkan validasi empiris dan perwujudan operasional pertama dari filosofi TISE-Valorize. Riset ini mengajukan pertanyaan utama: “Apakah sistem manajemen pengetahuan kolaboratif yang dirancang melalui prinsip TISE-Valorize, yang menggabungkan pemodelan kognitif formal dan alur kerja profesional, dapat meningkatkan pembentukan identitas profesional dan penciptaan nilai yang terukur dalam pendidikan teknik?” Langkah operasionalisasinya dibagi menjadi tiga tahap:]

1. **Mengoperasionalkan TISE:** Menghubungkan prinsip abstrak (*Homocordium*) dengan praktik terukur melalui mesin *PUDAL & PSKVE*.
2. **Meningkatkan Fidelitas Kognitif:** Menggunakan *Multi-Graph Taxonomy* untuk memodelkan proses berpikir ahli secara eksplisit.
3. **Menanamkan Alur Kerja Profesional:** Mengintegrasikan *Agile/Scrum* dan *Git/GitHub* untuk menciptakan simulasi profesi yang otentik.

## 1.2 TUJUAN PENELITIAN

Tujuan dari penelitian ini adalah:

1. Merumuskan pergeseran fundamental tujuan pendidikan dari penguasaan konten teknis yang rentan otomatisasi menuju pembentukan karakter, pola pikir, dan penilaian profesional (*becoming*) melalui kerangka *Triune-Intelligence Smart Engineering* (TISE) dan VALORIZE.
2. Mengembangkan dan memvalidasi sebuah purwarupa operasional bernama *Collaborative Knowledge Management System-Smart Engineering* (CKMS-SE) yang mengintegrasikan *Cognitive Load Theory* (CLT) dan *Self-Determination Theory* (SDT) ke dalam praktik kelas harian.
3. Mengukur efektivitas CKMS-SE terhadap tingkat motivasi intrinsik, identitas profesional, kapabilitas pembelajaran transfer, dan penciptaan nilai holistik (melalui pengukuran *PSKVE Value Portfolio*) jika dibandingkan dengan metode pengajaran konvensional.

## **BAB 2 METODOLOGI**

### **2.1 Umum**

Riset ini mengadopsi desain studi kuasi-eksperimental dengan grup kontrol ( $n = 71$ ) dan grup intervensi ( $n = 71$ ), yang dilakukan selama 12 minggu masa perkuliahan. Populasi terdiri dari 142 mahasiswa sarjana tingkat tiga di Institut Teknologi Bandung yang terdaftar dalam dua mata kuliah inti paralel, yakni Probabilitas & Statistik serta Sistem Teknik.

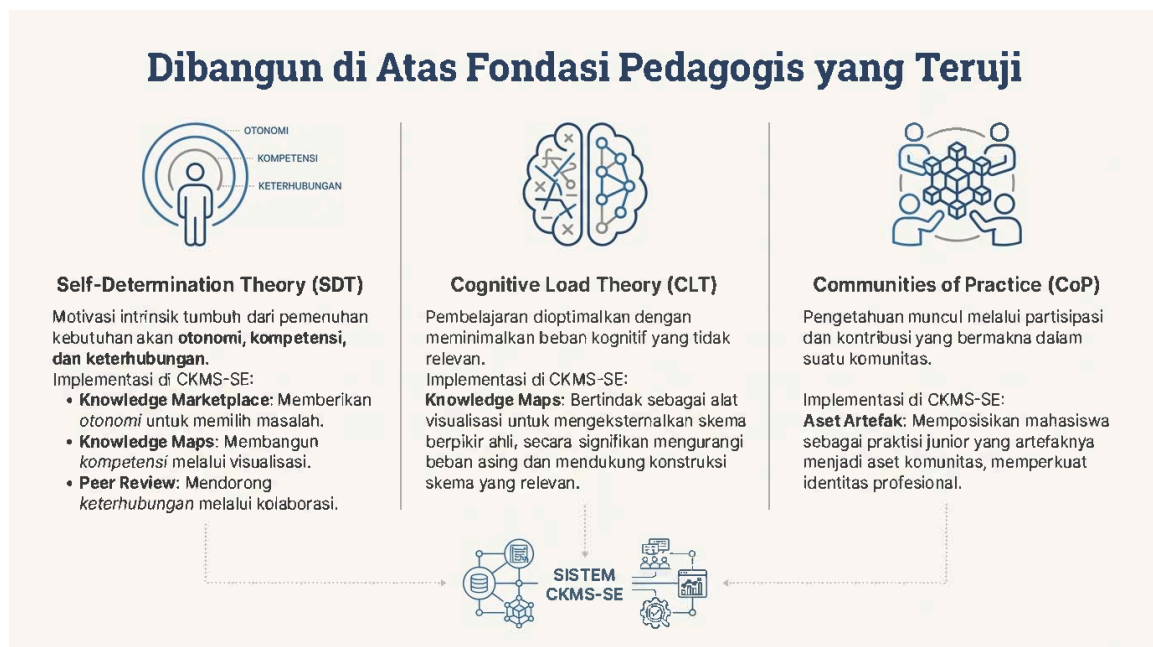


Figure 4: Dibangun di Atas Fondasi Pedagogis yang Teruji

Sistem CKMS-SE diimplementasikan berdasarkan tiga landasan teoritis utama:

1. **Self-Determination Theory (SDT):** Motivasi intrinsik tumbuh dari pemenuhan kebutuhan dasar akan otonomi, kompetensi, dan keterhubungan. Implementasinya di CKMS-SE adalah melalui *Knowledge Marketplace* (memberikan otonomi memilih masalah), *Knowledge Maps* (membangun kompetensi visualisasi), dan *Peer Review* (mendorong keterhubungan kolaboratif).
2. **Cognitive Load Theory (CLT):** Pembelajaran dioptimalkan dengan meminimalkan beban kognitif yang tidak relevan (ekstrinsik). Implementasinya menggunakan *Knowledge Maps* yang bertindak sebagai alat visualisasi untuk mengeksternalkan skema berpikir ahli, sehingga secara signifikan mengurangi beban asing dan mendukung konstruksi skema yang relevan di memori kerja.
3. **Communities of Practice (CoP):** Pengetahuan sejati muncul melalui partisipasi dan kontribusi bermakna di dalam komunitas. Implementasinya adalah melalui penciptaan “Aset Artefak”, yang memposisikan mahasiswa sebagai praktisi junior yang artefaknya dikelola menjadi aset komunitas, demi memperkuat identitas profesional mereka.

Instrumen pengukuran yang digunakan meliputi pengujian praktek untuk *Transfer Learning* (kemampuan penalaran pada konteks baru) serta kuesioner tervalidasi: *Intrinsic Motivation*



*Inventory* ( $\alpha = 0.89$ ) dan *Learning & Professional Identity in Practice Scale* (LPIPS,  $\alpha = 0.84$ ).

Grup intervensi mendapatkan perlakuan pengajaran berbasis arsitektur **CKMS-SE**, sebuah *Simulasi Profesi* berfidelitas tinggi yang bertumpu pada pendekatan kolaboratif berbasis Agile/Scrum dan kontrol versi menggunakan Git/GitHub.

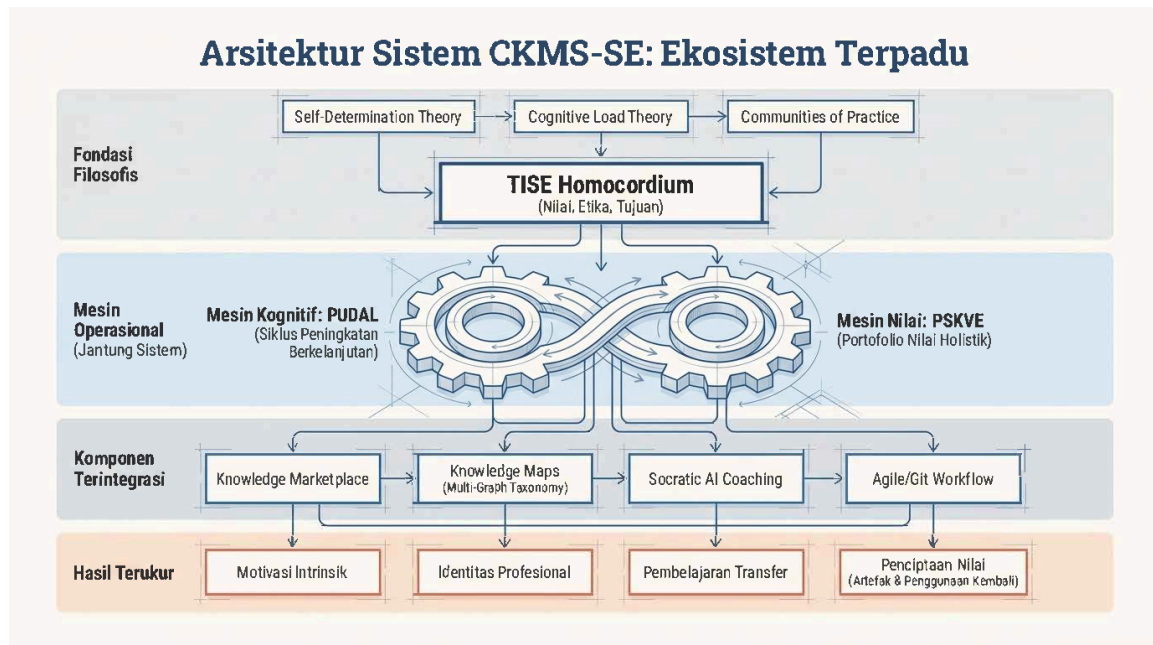


Figure 5: Arsitektur Sistem CKMS-SE: Ekosistem Terpadu

Secara teknis, ekosistem CKMS-SE digerakkan oleh dua mesin utama:

1. **Mesin PUDAL (Perceive, Understand, Decision-making, Act, Learning):** Sebuah siklus kognitif iteratif berbasis *Multi-Graph Taxonomy* untuk menuntun mahasiswa melatih penalaran kepakaran mereka di bawah pengawasan asisten kecerdasan buatan (*Socratic AI*).
2. **Mesin Portofolio Nilai (PSKVE):** Sebuah *Knowledge Marketplace* di mana tugas mahasiswa bertransformasi dari sekadar kewajiban ujian menjadi aset pengetahuan (Produk, Layanan, Pengetahuan, Nilai Berkelanjutan, dan Lingkungan) yang diberi *reward* simulasi ekonomi riil.

## 2.2 Mesin Kognitif PUDAL

### Mesin Kognitif: Siklus PUDAL untuk Penilaian Ahli

Inti kognitif sistem adalah siklus **PUDAL** yang berulang: **Perceive - Understand - Decision-making - Act - Learning**. Ini berfungsi sebagai "mesin mikro-evolusioner" untuk penyempurnaan berkelanjutan.

Pada setiap fase PUDAL, mahasiswa diharuskan menganalisis masalah melalui tiga lensa **Triune Intelligence (TI) TISE**.

Sistem ini secara formal menanamkan pemikiran holistik, memaksa sintesis antara nilai-nilai manusia, logika mesin, dan batasan alam.

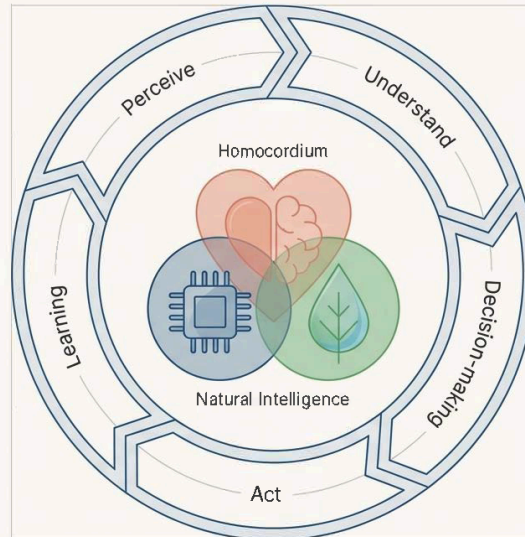


Figure 6: Mesin Kognitif: Siklus PUDAL untuk Penilaian Ahli

Riset ini menjelaskan bahwa inti kognitif dari sistem ini adalah siklus **PUDAL** yang berulang secara kontinu: **Perceive - Understand - Decision-making - Act - Learning**. Siklus ini berfungsi sebagai “mesin mikro-evolusioner” untuk penyempurnaan pengetahuan secara berkelanjutan. Pada setiap fasenya, mahasiswa diharuskan menganalisis masalah melalui tiga lensa **Triune Intelligence (TI) TISE**, yang divisualisasikan dengan tiga simbol di tengah lingkaran PUDAL:

- **Homocordium** (simbol hati/otak): Nilai-nilai kemanusiaan & etika.
- **Homodeus** (simbol *chip* cerdas): Logika komputasi & Kecerdasan Buatan (AI).
- **Natural Intelligence** (simbol daun): Hukum fisika dan batasan alam. Sistem ini secara formal menanamkan pemikiran holistik, yang memaksa sintesis antara nilai-nilai manusia, logika mesin, dan batasan ekologis alam.

## 2.3 Penerapan Artefak Kognitif



Figure 7: Dari Proses ke Produk: Artefak Kognitif Tingkat Tinggi

Siklus PUDAL bukan sekadar latihan teoritis; siklus ini menuntut mahasiswa memproduksi artefak kognitif spesifik yang sesuai dengan pemikiran Taksonomi Bloom tingkat tinggi, yang dimodelkan menggunakan kerangka *Multi-Graph Taxonomy* formal. Pemetaan fase dan artefaknya adalah sebagai berikut:

- **Perceive:** Menghasilkan *Factual Association Map*.
- **Understand:** Menghasilkan *Directed Acyclic Graph (DAG)* dan *Decomposition Tree*.
- **Decision-making:** Menghasilkan *Trade-Off Analysis Graph (Weighted Graph)*.
- **Act:** Menghasilkan *Process Application Map (Flowchart)*.
- **Learning:** Menghasilkan *Synthesized Design Map (Network Graph)*. Tujuannya adalah agar mahasiswa tidak hanya sekadar menyelesaikan masalah, melainkan secara eksplisit memodelkan dan mendokumentasikan proses berpikir ahli mereka.

## 2.4 Portofolio Nilai PSKVE



Figure 8: Mesin Nilai: Portofolio PSKVE untuk Penciptaan Nilai Holistik

Untuk menyelaraskan insentif dengan filosofi “Menjadi”, sistem merekayasa ulang sistem penilaian di dalam *marketplace*. Mata uang yang sebelumnya berbasis pada Taksonomi Bloom digantikan dengan **Portofolio Nilai Multi-Aset PSKVE**. Sistem ini memberikan penghargaan kepada mahasiswa karena telah menghasilkan artefak yang menciptakan “nilai holistik”, yang secara langsung menyimulasikan tiga pilar nilai rekayasa modern: Ekonomi, Perlindungan, dan Sosial. Dimensi portofolio tersebut terdiri dari:

- **P: Product** (Produk)
- **S: Service** (Layanan)
- **K: Knowledge** (Pengetahuan)
- **V: Value** (Nilai/Pertukaran)
- **E: Environment** (Lingkungan/Keberlanjutan)

## 2.5 Dimensi Nilai PSKVE

| Dimensi Nilai PSKVE dalam Praktik                                                                 |                           |                                   |                                                                           |
|---------------------------------------------------------------------------------------------------|---------------------------|-----------------------------------|---------------------------------------------------------------------------|
| Dimensi                                                                                           | Aset yang Diberikan       | Dimensi Nilai yang Dihargai       | Contoh Penciptaan Nilai oleh Mahasiswa                                    |
|  (K) Knowledge   | Knowledge Credits (KC)    | Kedalaman Kognitif                | Menghasilkan DAG formal yang memetakan hubungan konseptual yang kompleks. |
| (P) Product      | Productivity Units (PU)   | Nilai Ekonomi / Efisiensi         | Menulis skrip Python untuk mengotomatiskan perhitungan.                   |
|  (S) Service     | Service Vouchers (SV)     | Desain yang Berpusat pada Manusia | Membuat analisis pemangku kepentingan yang memprioritaskan aksesibilitas. |
| (V) Value        | Economic Shares (ES)      | Pertukaran Berkelanjutan          | Mengusulkan model bisnis yang layak untuk distribusi nilai yang adil.     |
|  (E) Environment | Sustainability Bonds (SB) | Nilai Perlindungan / Etika        | Melakukan Penilaian Siklus Hidup (LCA) untuk mitigasi dampak lingkungan.  |

Figure 9: Dimensi Nilai PSKVE dalam Praktik

Tabel yang merincikan aset yang diberikan dalam sistem dan contoh konkret penciptaan nilainya oleh mahasiswa:

- **(K) Knowledge:** Aset berupa *Knowledge Credits (KC)*. Menghargai Kedalaman Kognitif. Contoh: Menghasilkan DAG formal yang memetakan hubungan konseptual yang kompleks.
- **(P) Product:** Aset berupa *Productivity Units (PU)*. Menghargai Nilai Ekonomi / Efisiensi. Contoh: Menulis skrip Python untuk mengotomatiskan perhitungan.
- **(S) Service:** Aset berupa *Service Vouchers (SV)*. Menghargai Desain yang Berpusat pada Manusia. Contoh: Membuat analisis pemangku kepentingan yang memprioritaskan aksesibilitas.
- **(V) Value:** Aset berupa *Economic Shares (ES)*. Menghargai Pertukaran Berkelanjutan. Contoh: Mengusulkan model bisnis yang layak untuk distribusi nilai yang adil.
- **(E) Environment:** Aset berupa *Sustainability Bonds (SB)*. Menghargai Nilai Perlindungan / Etika. Contoh: Melakukan Penilaian Siklus Hidup (LCA) untuk mitigasi dampak lingkungan.

## 2.6 Implementasi Alur Kerja

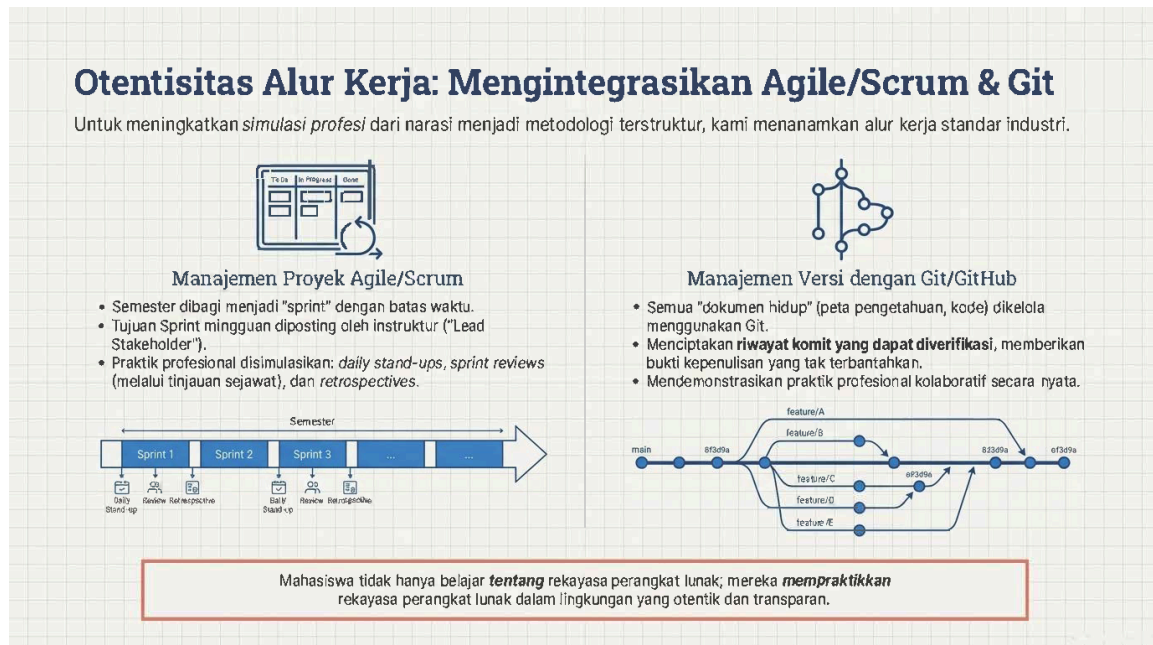


Figure 10: Otentisitas Alur Kerja: Mengintegrasikan Agile/Scrum & Git

Untuk meningkatkan simulasi profesi dari sekadar cerita menjadi metodologi terstruktur, CKMS-SE menanamkan standar alur kerja industri perangkat lunak. \* **Manajemen Proyek Agile/Scrum:** Semester perkuliahan dibagi menjadi "*sprint*" dengan batasan waktu tertentu. Tujuan *Sprint* mingguan akan diposting oleh instruktur (yang bertindak sebagai "Lead Stakeholder"). Mahasiswa melakukan praktik profesional yang disimulasikan seperti *daily stand-ups*, *sprint reviews* (berupa tinjauan sejawat), dan *retrospectives*.

- **Manajemen Versi dengan Git/GitHub:** Semua "dokumen hidup" seperti peta pengetahuan dan baris kode dikelola menggunakan Git. Ini menciptakan riwayat *commit* (perubahan) yang dapat diverifikasi, memberikan bukti kepenulisan yang tak terbantahkan, serta mendemonstrasikan praktik kolaboratif yang nyata melalui percabangan fitur (*feature branch*). Hasil akhirnya: Mahasiswa tidak hanya belajar *tentang* rekayasa perangkat lunak, mereka **mempraktikkan** rekayasa perangkat lunak secara langsung dalam lingkungan yang transparan.



## 2.7 Metodologi Penelitian



Figure 11: Validasi Kerangka Kerja: Desain Studi Eksperimental Kuasi

Gambar ini menguraikan desain studi untuk mengevaluasi dampak CKMS-SE selama satu semester (12 minggu).

- **Pembagian Kelompok / Populasi:** Melibatkan total **142 mahasiswa** teknik tahun ketiga di ITB. Dibagi menjadi Kelompok Intervensi (n=71) yang menggunakan sistem CKMS-SE dalam simulasi profesi (dosen bertindak sebagai *System Architect*), dan Kelompok Kontrol (n=71) yang mengikuti kurikulum standar konvensional.
- **Konteks Kursus:** Dilakukan di dua mata kuliah inti paralel, yakni Probabilitas & Statistik dan Sistem Teknik (dipilih karena penekanannya pada pemodelan sistem dan data yang merupakan pusat paradigma TISE).
- **Instrumen Pengukuran:** Meliputi domain Homocordium/Afektif (menggunakan *Intrinsic Motivation Inventory*,  $\alpha = 0.89$ , dan LPIPS,  $\alpha = 0.84$ ), domain Homodeus/Kognitif (Penilaian pembelajaran transfer, kesepakatan penilai  $\kappa = 0.84$ ), dan Penciptaan Nilai PSKVE (berdasarkan metrik analitik *platform*).

## BAB 3 HASIL DAN LUARAN YANG DICAPAI

### 3.1 HASIL PENELITIAN

Berdasarkan analisis performa terstruktur, studi membuktikan efikasi luar biasa dari kerangka kerja TISE-Valorize. Kelompok intervensi mencatat skor rata-rata sangat baik ( $M = 94.98$ ,  $SD = 7.21$ ) dibanding grup kontrol ( $M = 88.70$  dan  $86.54$ ) dengan tingkat dispersi/ variasi nilai yang jauh lebih kecil (Varians grup eksperimen =  $52.01$  berbanding  $>290$  pada grup kontrol). Ini membuktikan kerangka pembelajaran berhasil menciptakan kesetaraan pembelajaran dengan menekan *extraneous cognitive load*.

Lebih jauh, analisis instrumen pada kompetensi inti profesional memperlihatkan dampak transformatif.



Figure 12: Hasil: Dampak Transformatif pada Pembentukan Profesional

Secara spesifik, diukur menggunakan pengkategorian ukuran efek (Cohen's  $d$ ): \* **Motivasi Intrinsik:** Peningkatan sangat luar biasa ( $d = 0.97$ ,  $p < .001$ ). \* **Identitas Profesional:** Peningkatan yang signifikan ke arah pola pikir rekayasawan berintegritas ( $d = 0.84$ ). \* **Pembelajaran Transfer:** Lonjakan tajam dalam menyelesaikan kendala yang sama sekali belum pernah diajarkan di kelas ( $d = 0.95$ ).





Figure 13: Membangun Aset Pengetahuan Komunitas yang Berkelanjutan

Dalam aspek validasi *Value Co-Creation*, inisiatif *Knowledge Marketplace* telah mem- buahkan 427 artefak bernilai yang secara aktif diciptakan mahasiswa hanya dalam satu semester. Tingkat kolaborasi ditunjukkan dengan tingginya *Peer Review* (rata-rata 9.2 tin- jauan per artefak), di mana 16% dari pengetahuan dan aset komputasional yang diproduksi telah digunakan ulang secara nyata oleh angkatan (*cohort*) mahasiswa setelahnya.

### 3.2 LUARAN YANG DICAPAI

Sesuai dengan target, kemajuan riset (per April 2026) telah menghasilkan luaran-luaran konkret sebagai berikut:

1. **Konsep Teori (“Teori.doc”)**: Dokumen teori yang mengelaborasi desain *Knowledge Maps* mahasiswa dengan mensintesis *Cognitive Load Theory* (CLT) dan *Dual Coding Theory* (DCT), menunjukkan bahwa konversi bahasa sistem dan sinyal ke visual multi- dimensi bisa membebaskan kapasitas memori kerja (*working memory*).
2. **Artikel Jurnal Internasional (Status: *Under Review*)**: Draf jurnal berjudul “*Value Co-Creation Through Collaborative Knowledge Management in Smart Engineering Education: An Empirical Validation of the TISE-Valorize Framework*” (atau dengan variasi “*Orchestrating Value Co-Creation...*”) telah dikirimkan ke jurnal **Frontiers**

in **Education** yang merupakan jurnal internasional terkemuka (Quartile 2 Scopus).  
*Acknowledgment submission* dan draf terlampir pada bagian Lampiran.

## BAB 4 KESIMPULAN DAN SARAN

### 4.1 KESIMPULAN

Pelaksanaan sistem CKMS-SE membuktikan secara empiris bahwa filosofi TISE-Valorize dapat dieksekusi sebagai alat transformatif. Riset ini berhasil memvalidasi bahwa melalui pengelolaan manajemen beban kognitif (CLT) dengan visualisasi skema *Multi-Graph* serta stimulasi kebutuhan dasar otonomi (SDT), tujuan pendidikan teknik dapat digeser dari yang tadinya rentan diotomatisasi AI (hanya menguasai algoritma/rumus) menjadi kompetensi yang bertumpu pada pembangunan karakter etis, kreatif, dan kepakaran (*Homocordium*) yang mustahil ditiru oleh mesin. Lonjakan yang signifikan pada nilai motivasi intrinsik dan keseragaman capaian belajar mengkonfirmasi bahwa mahasiswa tidak hanya belajar teknik, tetapi belajar untuk *menjadi* insinyur sejati.



Figure 14: Paradigma Baru: Membentuk Insinyur yang Tangguh di Era AI

Sebagai penutup, gambar ini merangkum perbandingan paradigma “Pendidikan Teknik Kemarin” yang berfokus pada apa yang mahasiswa “Ketahui”, rentan diautomasi, dan sekadar mencetak kompetensi teknis dasar. Ini dibandingkan dengan “Pendidikan Teknik

Besok (TISE-Valorize)” yang menggeser fokus pada “Menjadi” seorang insinyur profesional. Model ini membina pembentukan profesional holistik yang nilainya berlabuh pada **Homocordium**: kapabilitas manusia yang unik. Kesimpulannya: CKMS-SE membuktikan bahwa kita dapat merancang ekosistem pembelajaran yang secara sistematis menumbuhkan penilaian ahli, penalaran etis, dan identitas profesional. Hal ini bukan tentang bersaing melawan AI, melainkan tentang **mengembangkan kualitas-kualitas manusia yang tidak dapat ditiru oleh AI**. Inilah masa depan pendidikan teknik yang berkelanjutan dan bernilai di era masa depan.

## 4.2 SARAN

Meski menghasilkan luaran yang impresif, riset saat ini dilakukan secara spesifik pada populasi terbatas (dua mata kuliah di satu institusi) dengan sampel  $N = 142$ . Ke depannya, disarankan agar riset diperluas pada skala multi-institusi dengan besar sampel yang disarankan minimum  $N \geq 200 - 250$  per grup untuk mencapai kekuatan statistik ideal (*statistical power*  $\geq 0.80$ ). Pengukuran yang lebih langsung terhadap beban kognitif (misalnya menggunakan NASA-TLX) juga patut disematkan pada fase riset mendatang.

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## Lampiran 1

### FORMULIR EVALUASI ATAS CAPAIAN LUARAN KEGIATAN

**Peneliti Utama:** : Prof. Ir. Armein Z.R. Langi, M.Sc., Ph.D.

---

**Perguruan Tinggi:** : Institut Teknologi Bandung

**Judul:** : Pengembangan dan Validasi Model Pembelajaran VAL-ORIZE: Kerangka Pedagogis Berbasis AI untuk Membentuk Identitas Profesional Rekayasawan di Era Digital

**Tahun Kegiatan:** : 2025 - 2026

#### TARGET PUBLIKASI INTERNASIONAL (JOINT PUBLICATION):

| No. | Nama Jurnal Internasional                    | Jumlah Artikel |
|-----|----------------------------------------------|----------------|
| 1.  | Frontiers in Education (Terindeks Scopus Q2) | 1              |

#### CAPAIAN SELURUH PUBLIKASI DARI PENELITI UTAMA DAN MITRA:

##### 1. PUBLIKASI JURNAL ILMIAH INTERNASIONAL TERINDEKS SCOPUS

ARTIKEL  
JURNAL  
KE-1

---

Nama jurnal : Frontiers in Education  
yang dituju

Klasifikasi : Jurnal Internasional Terindeks SCOPUS Q1/Q2 (Q2)  
jurnal

Judul artikel : Value Co-Creation Through Collaborative Knowledge Management in Smart Engineering Education: An Empirical Validation of the TISE-Valorize Framework

Status naskah : • Under Review

##### 2. KETERLIBATAN PENELITI UTAMA DAN MITRA PT PADA PUBLIKASI ARTIKEL ILMIAH 1:

- **Meliana Christianti Johan:** Penulis Pertama (First Author/Corresponding Author), berperan mengelola pengumpulan data eksperimen di lapangan serta menyusun manuskrip utama.
- **Armein Z. R. Langi & I Gusti Bagus Baskara Nugraha (ITB):** Mengawal pondasi desain Sistem Cerdas, konsep kerangka TISE, dan analisis metodologis berbasis teknologi informasi.
- **Radiant Victor Imbar & Parlindungan Sinaga:** Mendukung tinjauan edukasi teknologi komprehensif serta penyelarasan pedagogi dengan aspek TISE Homocordium.

Bandung, 7 April 2026

Peneliti Utama,



**Prof. Ir. Armein Z.R. Langi, M.Sc., Ph.D.**

NIP. 196208171990031002

## Lampiran 2.

### Ringkasan Laporan Kemajuan Program Riset Kolaborasi Indonesia-EQUITY

World Class University (WCU) 2025-2026

| Komponen                          | Penjelasan                                                                                                                                                                                                                                                                                                                          |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Judul Riset</b>                | Pengembangan dan Validasi Model Pembelajaran VALORIZE: Kerangka Pedagogis Berbasis AI untuk Membentuk Identitas Profesional Rekayasawan di Era Digital                                                                                                                                                                              |
| <b>Peneliti Utama</b>             | Prof. Ir. Armein Z.R. Langi, M.Sc., Ph.D.                                                                                                                                                                                                                                                                                           |
| <b>Peneliti Mitra</b>             | Prof. Dr. Dr. Parlindungan Sinaga, M.Si                                                                                                                                                                                                                                                                                             |
| <b>Perguruan Tinggi</b>           | Institut Teknologi Bandung & Universitas Pendidikan Indonesia                                                                                                                                                                                                                                                                       |
| <b>Target Capaian</b>             | Pertengahan riset: Memformulasikan landasan teoritis konseptual TISE-Valorize dalam mengelola memori kerja. Akhir riset: Publikasi luaran <i>framework</i> di jurnal bereputasi tinggi yang memvalidasi pergeseran performa mahasiswa secara empiris.                                                                               |
| <b>Hasil yang sudah diperoleh</b> | <ul style="list-style-type: none"><li>• Tersusunnya laporan kajian pustaka psikologis kognitif (Teori.doc).- Terukurnya performa uji 142 mahasiswa yang membuktikan efektivitas sistem CKMS-SE pada 3 domain indikator.- Tersusunnya naskah manuskrip Frontiers.pdf yang sudah dikirim (<i>submitted / under review</i>).</li></ul> |
| <b>Persentase</b>                 | Diperkirakan kemajuan hingga tahap laporan ini sudah mencapai kisaran 80-85% (data sudah disintesis seluruhnya).                                                                                                                                                                                                                    |

| Komponen                          | Penjelasan                                                                                                                                                                                                                                                |
|-----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Manajemen Riset Kolaborasi</b> | Koordinasi integratif melalui repositori Git dan sinkronisasi berkala antara pembina teknologi di ITB dan unit pendidikan.                                                                                                                                |
| <b>Pendanaan</b>                  | Tidak ada hambatan spesifik, dana berjalan lancar dalam pengelolaan kegiatan penelitian dan penyusunan jurnal.                                                                                                                                            |
| <b>Kelanjutan riset</b>           | Pemantauan progres jurnal <i>Under Review</i> ke fase perbaikan ( <i>revision</i> ) lalu tahapan penerbitan ( <i>published</i> ). Serta merekomendasikan pelebaran purwarupa dengan besar sampel ke institusi lain.                                       |
| <b>Hambatan dan kesulitan</b>     | Diperlukan persiapan <i>platform</i> dengan daya analisis <i>Socratic AI</i> berskala lebih besar yang menuntut kolaborasi komputasional. Evaluasi jangka panjang (kestabilan retensi ingatan per tahun) belum dapat dicakup di batas waktu semester ini. |

Bandung, 7 April 2026

Peneliti Utama,



**Prof. Ir. Armein Z.R. Langi, M.Sc., Ph.D.**

NIP. 196208171990031002



# Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

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## Scope Statement

This paper presents the Triune-Intelligence Smart Engineering-VALORIZE (TISE-VALORIZE) Framework, which brings together human, artificial, and natural intelligence in the Perceive-Understand-Design-Act-Learn (PUDAL) cycle and a set of value engines (PSKVE) for digital engineering education. An experimental investigation including 138 students demonstrated a significant rise in their academic scores ( $M=94.98$ ), accompanied with a reduction in the score range by as much as 84% in comparison to the control group. This suggests that it was more fair and useful to learn. Value Co-Creation, collaborative communities of practice, and AI integration are all in line with the digital learning innovation trends that Frontiers in Education sees, especially in the Digital Learning Innovations area. This research may add to the body of work on AI-based teaching and shared values by providing empirical data and a solid theoretical background. This will help the digitalization agenda of higher education.

## Conflict of interest statement

**The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest**

## Credit Author Statement

**Armein Z. R. Langi:** Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing. **I Gusti Bagus Baskara Nugraha:** Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – review & editing. **Meliana Christianti Johan:** Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Project administration, Resources, Validation, Visualization, Writing – original draft, Writing – review & editing. **Parlindungan Sinaga:** Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Visualization, Writing – original draft. **Radiant Victor Imbar:** Conceptualization, Data curation, Formal Analysis, Methodology, Resources, Supervision, Validation, Visualization, Writing – review & editing.

## Keywords

artificial intelligence, Cognitive Load Theory, Engineering Education, Professional identity formation, self-determination theory, TISE-VALORIZE framework, value co-creation

## Abstract

Word count: 232

AI can automate technical activities, but it can't teach skills that only people have. This study proposes the TISE-VALORIZE framework and presents a preliminary empirical examination of its association with student performance outcomes in digital engineering education. A posttest-only quasi-experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering students: one intervention group implementing TISE-VALORIZE ( $n = 53$ ) and two control groups receiving conventional instruction ( $n = 42$ ;  $n = 43$ ). Student performance was evaluated through Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention group showed higher mean scores ( $M = 94.98$ ,  $SD = 7.21$ ) and lower score variance ( $Var = 52.01$ ) than Control 1 ( $M = 88.70$ ,  $SD = 18.33$ ,  $Var = 335.78$ ) and Control 2 ( $M = 86.54$ ,  $SD = 17.05$ ,  $Var = 290.98$ ). A one-way ANOVA indicated a significant group difference,  $F(2, 135) = 4.41$ ,  $p = .014$ ,  $\eta^2 = .061$ , and Tukey post-hoc analysis showed a significant difference between the intervention group and Control 2, while the difference between the intervention group and Control 1 did not reach significance. These preliminary results suggest that the framework may support more consistent learning outcomes. However, direct measures of cognitive load, motivation, and related psychological processes were not included in the present study. These findings provide preliminary support for the framework's potential to improve performance consistency, while larger-scale studies with direct measures of cognitive and motivational processes are needed.

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## ***Ethics statements***

### ***Studies involving animal subjects***

Generated Statement: No animal studies are presented in this manuscript.

### ***Studies involving human subjects***

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### ***Inclusion of identifiable human data***

Generated Statement: No potentially identifiable images or data are presented in this study.

### ***Data availability statement***

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### ***Generative AI disclosure***

The author(s) verify and take full responsibility for the use of generative AI in the preparation of this manuscript. Generative AI was used

In this study, we used Canva as a drawing application to create Figure 1 and also used Grammarly to check the grammar and writing of the manuscript..

# Orchestrating Value Co-Creation in Digital Education Using the TISE-VALORIZE Framework: From Knowing to Becoming

1 Meliana Christianti Johan<sup>1,2\*</sup>, Armein Z. R. Langi<sup>1</sup>, I Gusti Bagus Baskara Nugraha<sup>1</sup>, Radiant  
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10 **Keywords:** artificial intelligence, cognitive load theory, engineering education, professional  
11 identity formation, self-determination theory, TISE-VALORIZE framework, value co-creation

12 **Abstract**

13 AI can automate technical activities, but it can't teach skills that only people have. This study proposes  
14 the TISE-VALORIZE framework and presents a preliminary empirical examination of its association  
15 with student performance outcomes in digital engineering education. A posttest-only quasi-  
16 experimental study with non-equivalent cross-cohort groups involved 138 undergraduate engineering  
17 students: one intervention group implementing TISE-VALORIZE ( $n = 53$ ) and two control groups  
18 receiving conventional instruction ( $n = 42$ ;  $n = 43$ ). Student performance was evaluated through  
19 Structured Academic Activities assessed with Bloom's Taxonomy-aligned rubric. The intervention  
20 group showed higher mean scores ( $M = 94.98$ ,  $SD = 7.21$ ) and lower score variance ( $Var = 52.01$ ) than  
21 Control 1 ( $M = 88.70$ ,  $SD = 18.33$ ,  $Var = 335.78$ ) and Control 2 ( $M = 86.54$ ,  $SD = 17.05$ ,  $Var = 290.98$ ).  
22 A one-way ANOVA indicated a significant group difference,  $F(2, 135) = 4.41$ ,  $p = .014$ ,  $\eta^2 = .061$ , and  
23 Tukey post-hoc analysis showed a significant difference between the intervention group and Control  
24 2, while the difference between the intervention group and Control 1 did not reach significance. These  
25 preliminary results suggest that the framework may support more consistent learning outcomes.  
26 However, direct measures of cognitive load, motivation, and related psychological processes were not  
27 included in the present study. These findings provide preliminary support for the framework's potential  
28 to improve performance consistency, while larger-scale studies with direct measures of cognitive and  
29 motivational processes are needed.

30 **1 Introduction**

31 Engineering education is shifting from a content-based approach to competency development  
32 (Charytanowicz, 2023). Other empirical studies indicate that the disparity between education and  
33 industry endures, even among high-achieving graduates: analyses of software engineering practices  
34 consistently reveal a misalignment between the focus of higher education programs and the  
35 requirements of practitioners in professional settings, including both technical and professional aspects  
36 (Garousi, Giray and Tuzun, 2019). To build a long-lasting smart campus ecosystem, many tools and

frameworks are being created to test preparedness and include digital transformation (Imbar *et al.*, 2025). Engineering education has long emphasized foundational knowledge, routine problem solving, and performance assessment. Students study the basics, figure out how to tackle frequent situations, and test their skills with tests. High grades showed that you were ready to work in the field. But this approach included an unspoken assumption: that the mental effort needed to solve engineering issues was the same as the mental labor needed to be an engineer. This assumption is no longer valid.

Learning requires learners to organize information meaningfully and integrate it into schemas stored in long-term memory (Sweller, Van Merriënboer and Paas, 1998). One important thing about working memory, which is a mental region where new knowledge is kept, is that it can only contain roughly 3–7 bits of information at a time. Unless actively rehearsed, such information typically decays within 15–30 seconds (Miller, 1956). Students will have trouble learning and developing their abilities if the environment has too much information in it. Cognitive Load Theory (CLT) offers a solution: instructional design should minimize extraneous cognitive load to optimize working memory for effective learning termed 'germane load' by CLT researchers. CLT serves as a cognitive framework for crafting educational experiences that cultivate robust skills (Robins, 2022). It must alleviate superfluous cognitive burden to liberate working memory for authentic learning, and it must structure teaching to facilitate the development of meaningful, practical knowledge frameworks that embody professional competence. It must also do this in a manner that respects autonomy, competence, and relatedness that keep people interested. CLT posits that the complexity of a topic correlates with the interaction of its components. Darejeh, Marcus and Sweller, (2021) examined the impact of narrative style on cognitive load during the acquisition of a difficult productivity software program. The researchers used two categories of materials, distinguished by low and high element interaction, according to each story style.

The theoretical framework delineated below aims to actualize this possibility. It is based on well-known learning theories (Cognitive Load Theory (CLT), Self-Determination Theory (SDT), and research on expertise) and considers the realities of modern technology (AI integration, digital tools, and participatory design). Its goal is to create professionals who can lead and make a difference in a world that is changing quickly. In this preliminary study, we empirically evaluate only the performance-related dimension of the framework, while the motivational, cognitive-load, transfer, and professional-identity mechanisms remain theoretical propositions requiring direct measurement in future work. The framework draws on Cognitive Load Theory, Self-Determination Theory, and professional identity formation because these perspectives address three complementary dimensions of engineering education. Cognitive Load Theory explains how structured learning design and external representations may support schema construction and reduce unnecessary processing demands. Self-Determination Theory explains how autonomy, competence, and relatedness may support sustained engagement in learning activities. Professional identity formation helps explain why professional simulation and value-oriented participation matter in engineering education beyond content mastery alone. Together, these perspectives provide the theoretical basis for the TISE-VALORIZE framework and for its expected influence on both learning processes and performance outcomes.

In the present preliminary empirical study, two hypotheses were examined directly:

- H1. Students in the TISE-VALORIZE group will achieve higher Structured Academic Activities scores than students in the conventional instruction groups.
- H2. Students in the TISE-VALORIZE group will show lower score dispersion than students in the conventional instruction groups.

82 In addition, the framework proposes several secondary hypotheses for future empirical testing:

- 83 • H3. The framework may reduce perceived cognitive load through structured knowledge  
84 mapping and scaffolding.
- 85 • H4. The framework may support intrinsic motivation and learning agency through autonomy-  
86 supportive and contribution-oriented learning activities.
- 87 • H5. The framework may support stronger professional identity formation and learning transfer  
88 through professional simulation and value-oriented participation.
- 89 • H6. The framework may increase the quality and reuse value of student-produced artefacts  
90 within the learning community.

91 Because the present study did not include direct measures of cognitive load, motivation, professional  
92 identity, or transfer, only H1 and H2 are evaluated empirically in this paper.

## 93 **2 Materials and Methods**

### 94 **2.1 Theoretical Foundations: Merging Cognitive Science and Pedagogy**

#### 95 **2.1.1 Fundamental Ideas About Education**

96 The framework is based on several interconnected ideas that are meant to meet the needs of the AI  
97 future. First, Value-Oriented Education (VOE). In traditional education, teachers teach technical  
98 information, and students learn it. VOE alters this goal. Higher education should teach how to be good  
99 people, how to be professionals, and how to give back to society. Students learn how to develop code  
100 and why technological choices might hurt people or the environment, for example. Second, Value Co-  
101 Creation (VCC). This word originates from marketing and service writing and talks about how  
102 businesses and customers work together to produce value. In education, VCC means that students do  
103 not merely receive knowledge passively; they actively contribute to creating learning experiences that  
104 matter. They collaborate with educators and technology to develop resources such as knowledge maps,  
105 problem-solving tools, and educational aids for their classmates and future learners. In this way,  
106 students become co-creators instead than merely consumers of knowledge (Pralhad and Ramaswamy,  
107 2004; Grönroos and Voima, 2013; Robayo-Pinzon *et al.*, 2023)

108 Knowledge Management (KM) is a broad way for an organization to gather, develop, protect, share,  
109 and use knowledge gained from its experiences (Mohammed, 2021; Deev and Finogeev, 2023). It  
110 works well with the focus of VOE and VCC. Collaborative Learning (CL) is a strategy in which a small  
111 group of students work together to achieve a shared objective or complete an assignment (Ouyang, Xu  
112 and Cukurova, 2023; Ramadevi *et al.*, 2023). Lastly, Personalized Learning (PL) is a method of  
113 teaching that changes based on how fast students learn and how lessons are given to help them achieve  
114 their best (Labib, Canós and Penadés, 2017; Deev and Finogeev, 2023; Apirating, 2025).

#### 115 **2.1.2 Psychological and Cognitive Foundations**

116 CKMS-SE combines old ways of thinking about psychology and cognition:

- 117 • Self-Determination Theory (SDT) explains that individuals are intrinsically driven when their  
118 demands for autonomy, competence, and relatedness are satisfied (Weydner-volkmann and  
119 Bär, 2024; Annamalai and Nasor, 2025).

- Cognitive Load Theory (CLT) emphasizes that successful learning is facilitated by reducing irrelevant cognitive burden (extrinsic load) and enhancing cognitive load that aids the learning process (Alsaffar, Alfayly and Ali, 2022; Apirating, 2025).
- People that have similar aims, interests, and skill levels make up Communities of Practice (CoP). They acquire information collaboratively by active participation and contribution as a cohesive unit (Tsang and Tsui, 2017; Taft, 2019).

### 2.1.3 Self-Determination Theory and Intrinsic Motivation

CLT elucidates the cognitive frameworks integral to the learning process, while SDT emphasizes the motivating factors that sustain student engagement and facilitate profound learning (Ninghardjanti *et al.*, 2025). SDT says that human motivation exists on a scale. At one end is external regulation, where people do things because they feel pressure or get benefits from outside sources. At the other end is integrated regulation, where people do things that are in line with their own beliefs and identities. SDT also makes a distinction between the amount of motivation (how strongly someone feels motivated) and the kind of motivation (the qualities or categories of motivation they have).

Students who still use conventional learning approaches tend to depend on grades or test results as rewards. They usually use shallow learning tactics, stop when the benefits stop, and say they are less happy. Conversely, learners driven by intrinsic factors—such as curiosity, autonomy, and the aspiration to make a significant impact, employ more profound learning strategies, maintain persistence despite the lack of external incentives, and experience enhanced well-being and continuous engagement.

### 2.1.4 Cognitive Load Theory and Schema Construction

New information is first processed in working memory. Working memory has severe capacity and duration constraints (Duran, Zavgorodniaia and Sorva, 2022). Learning is the process of transferring information from working memory to long-term memory, where it is organized into mental schemas that integrate multiple elements into a coherent whole. The creation and automation of such schemas are central to expertise development (Robins, 2022).

This research established the VALORIZE strategy, which aims to design learning that alleviates the three forms of cognitive burden on students and enhances information processing in their working memory, drawing on cognitive load theory (CLT) and dual coding theory (DCT). Students convert the signal and system domains, which were initially expressed in words and arithmetic equations, into knowledge maps and problem-solving maps.

The following design principle is clear: to make learning as effective as possible, instructional designers should reduce extraneous load, manage intrinsic load correctly (by providing scaffolding for difficult material), and increase germane load, which is the amount of working memory used for productive learning activities (Duran, Zavgorodniaia and Sorva, 2022).

### 2.1.5 Community of Practice: Professional Identity and Expertise Development

The idea of professional identity, which is taking on the responsibilities, attitudes, and behaviors that come with a job, is an important part of the framework. Professional identity isn't just something you get by learning new things. It develops over time as you do real professional work, get advice from experienced professionals, and gradually adopt professional standards and values (Pischetola, 2021; Shal *et al.*, 2025).

Moreover, ethical thinking and responsible professional judgment are fundamental to professional identity (Große-Bölting *et al.*, 2023). Engineers do not work in a vacuum; their choices have an impact on people's health, the environment, and social justice (Cortázar *et al.*, 2024). A professional identity must include a dedication to ethical ideals (Szynkiewicz, 2025) and the ability to maneuver through intricate scenarios where technical and ethical factors converge (Johannsen, 2026).

## **2.2 The TISE Architecture: Operationalizing Multifaceted Intelligence**

In this paper, TISE-VALORIZE refers to the overarching theoretical framework, while CKMS-SE refers to its system-level implementation architecture. The suggested way to address the educational gap is the Collaborative Knowledge Management System for Smart Engineering (CKMS-SE). CKMS-SE turns the TISE-VALORIZE theory into a real way to solve problems that use human, artificial, and natural intelligence in students' everyday learning activities. The approach is based on dual-engine architecture. The first engine, called the PUDAL engine, organizes students' thinking into repeated cycles of Perceive–Understand–Design–Act–Learn (PUDAL). The second engine, called the PSKVE engine, manages a multi-asset value portfolio (Product, Service, Knowledge, Values, Environment) to find and reward multi-dimensional value co-creation.

### **2.2.1 Defining the Triune-Intelligence Model**

Triune-Intelligence Smart Engineering (TISE) proposes a method for intelligent problem-solving that integrates three complementary forms of intelligence: Human intelligence distinguishes individuals by their capacity for moral reasoning, alignment of values, strategic thinking, innovative solution generation, and comprehension of contextual nuances. Artificial intelligence is the capacity to swiftly see patterns, make things better, put together a lot of knowledge, and follow computer instructions that are clear. AI systems are fantastic at working with large amounts of data, seeing minor patterns, and repeating the same tasks again with remarkable precision. Natural Intelligence: The knowledge that comes from the rules of society, the way things work in nature, and the way people think they should be. This kind of intelligence involves learning about the rules and patterns of nature, how various civilizations make sense of things, and the information that is passed down via different groups and traditions. It makes sure that new technologies are good for people and the environment in the long run.

### **2.2.2 The PUDAL Thinking Cycle as A Cognitive Method**

The PUDAL engine has five steps that make up a full cycle: Perception (P): Students find stakeholders, data sources, and environmental limitations that are important to the issue. Understanding (U): They use the values at play, hidden patterns, and physical laws to get a full picture of what's going on. Solution Development (D): They look at more than one option, not simply the best one. Action (A): They put the selected solution into action by making a physical item, such code, an analytical report, a user interface, or a prototype. Learning (L): They think about how what they do affects people, computers, and the environment.

VALORIZE employs Knowledge Maps structured in a multi-graph taxonomy to make this expert cognitive process clear. This is like a "grammar of knowledge" that helps experts think. Formal knowledge artifacts back up each step of PUDAL (see Table 1). The whole process isn't a straight line; instead, the PUDAL cycle is iterative and always changing, like micro-evolution that leads to constant progress.



Table 1. PUDAL Cycle: Analysis of TISE and Artefact Requirements

| PUDAL Phase                      | Human Intelligence Analysis Task                                                                            | Artificial Intelligence Analysis Task                                                         | Nature Intelligence Analysis Task                                                              | Required VALORIZE Artefacts                                            |
|----------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Perceive (P)                     | Find out who the most important stakeholders are and what they think about the problem in general.          | Define data sources and collection strategies.                                                | Find the most important environmental indicators and physical limits that need to be assessed. | Factual Association Map                                                |
| Understand (U)                   | Find out what stakeholders' value, what ethical challenges they face, and what the cultural environment is. | Analyze data to identify patterns, correlations, and barriers.                                | Model the laws of physics and the rules of nature that control the system.                     | Portfolio: Directed Acyclic Graph (DAG), Decomposition Tree, Flowchart |
| Decision-making and Planning (D) | Set up ethical guardrails and success metrics that are based on people.                                     | Use optimization algorithms or simulations to evaluate solution options.                      | Look at your possibilities in light of your resources and how well they work.                  | Trade-Off Analysis Map (Weighted Graph)                                |
| Act-Response (A)                 | Make user interfaces and service delivery models that people can utilize.                                   | Make algorithms for controlling things, pipelines for processing data, or actions for robots. | Design physical components and artefact mechanisms.                                            | Process Application Map (Flowchart)                                    |
| Learning-Evaluating (L)          | Think about how the answer will affect people's health and equality in the long run.                        | Think about how the answer will affect people's health and equality in the long run.          | Evaluate the non-negotiable final impact of the solution on the environment.                   | Synthesized Design Map (Network Graph)                                 |

### 2.2.3 The PSKVE engine as a value-creation method

The Knowledge Marketplace was changed into a multi-asset PSKVE value portfolio (see Table 2).

Table 2. Dimensions of PSKVE and TISE-VALORIZE Assets

| PSKVE Dimension | TISE-VALORIZE Asset       | Dimension of Valued Worth                           | Example of Artefact Value Creation                                                                       |
|-----------------|---------------------------|-----------------------------------------------------|----------------------------------------------------------------------------------------------------------|
| Product (P)     | Productivity Units (PU)   | Economic Value (Productivity and Efficiency)        | Writing Python programs to make computations easier or improve designs.                                  |
| Service (S)     | Service Vouchers (SV)     | Human-Centered Design (Well-being and Empathy)      | Do a thorough study of stakeholders or a map of the user experience.                                     |
| Knowledge (K)   | Knowledge Credits (KC)    | Cognitive Depth (Intellectual Capacity)             | Makes a formal Directed Acyclic Graph (DAG) that shows how different concepts are related to each other. |
| Value (V)       | Economic Shares (ES)      | Sustainable Exchange (Financial/Social Viability)   | Come up with a workable business model or cost-benefit analysis for fair value distribution.             |
| Environment (E) | Sustainability Bonds (SB) | Protective/Ethical Values (Risk and Sustainability) | Do a life cycle evaluation or create systems that really reduce their influence on the environment.      |

## 2.3 Operationalization through the Knowledge Marketplace and Knowledge Maps

The method uses two main tools to embed the dual engines into daily learning: (1) a knowledge Marketplace, where lecturers post weekly "learning needs" as sprint goals and students voluntarily respond by making PSKVE-typed artefacts; and (2) AI-assisted Knowledge Maps will help students express their thoughts using the formal multi-graph taxonomy, like Directed Acyclic Graphs (DAGs), flowcharts, and weighted graphs. Git and GitHub make it simple to maintain track of all their assets and make sure that contributions are clear, can be used again, and are straightforward to verify.

## 214 **2.4 VALORIZE Learning: A Participatory Pedagogy for Professional Formation**

### 215 **2.4.1 Core Principles and the Change from Knowing to Being a Professional**

216 VALORIZE fundamentally shifts the focus of education from content mastery, which is learning and  
217 repeating facts and procedures, to professional formation, which is developing the intellectual, ethical,  
218 and social traits that make a person a good, responsible professional. This change isn't only a change  
219 in words. It has big effects on how teachers and students are responsible for their own learning, how  
220 they teach, and how they grade. In a content-based learning model, pupils are seen as individuals who  
221 acquire knowledge. The curriculum includes things that need to be taught, and the teachers are  
222 professionals who offer that knowledge. How effectively pupils can repeat that material is how success  
223 is measured. In a professional development model, on the other hand, students are seen as future  
224 professionals who are doing real work. The curriculum includes things that happen in the workplace,  
225 and teachers act as mentors to help students grow professionally. Success is based on the ability to add  
226 value and contribute.

### 227 **2.4.2 Professional Simulation as Pedagogical Strategy**

228 The lecturer's job varies a lot. The lecturer is no longer merely a "sage on the stage" who shares  
229 information. Now, they are "representative and role models of the profession." This isn't just a shift in  
230 words. There are real-world effects: the lecturer demonstrates how to handle difficulties like a pro by  
231 stating their ideas out loud and making their unspoken knowledge evident. The teacher teaches students  
232 how to be professionals by providing them feedback that goes beyond merely being accurate and looks  
233 at aspects like how well they match with professional standards, quality, and depth.  
234 Professional values are not only an additional feature of the VALORIZE paradigm; they are the most  
235 significant part of any learning activity. When students create control systems, they must not only  
236 improve their technical performance but also consider their impact on energy consumption, user-  
237 friendliness, and the environment. Learning sends a clear message: being good at something and being  
238 ethically responsible are two sides of the same coin.

## 239 **2.5 The Knowledge Marketplace: An Economic Model for Motivation and Value Creation**

240 The knowledge map is a mental structure that helps people learn, while the Knowledge Marketplace is  
241 a social and motivating framework. This marketplace changes the way students, teachers, and  
242 knowledge interact with each other. Instead of just following rules and taking in information, they are  
243 now creating and sharing it.

### 244 **2.5.1 How it works and why**

245 The Knowledge Marketplace functions like a real marketplace. Teachers publish "learning needs" as  
246 knowledge assignments (such as building a signal-system concept map), and students respond by  
247 generating high-quality artifacts, like knowledge maps or problem-solving tools. Then, students may  
248 use digital money to "buy" artifacts that match the requirements. These artifacts are enduring resources  
249 that future organizations may use to learn from and build on. This sets off a virtuous cycle: when  
250 students' effort adds value, they receive higher marks, and their classmates have access to richer  
251 learning resources.

252 Teachers need to ask for artifacts first. They normally put them into groups based on how hard they  
253 are to think about and the technical field. Then, students make Primitive Maps and Problem-Solving  
254 Models that are good for teaching. It takes a lot of time and thinking to become an expert at these.  
255 Next, the artifacts are evaluated to see whether they are comprehensive, clear, and in line with what

experts know. People who pass the examinations may "buy" them via a tiered digital currency system. The course's Knowledge Management System releases a few artifacts. These artifacts become permanent resources that present and future students may use and edit, which helps the community learn more. Lastly, pupils' final marks depend on how much money they gain by selling artifacts. This makes it clear that performing well in higher education is directly related to being able to undertake important intellectual work. It also promotes intrinsic motivation (since the activity is regarded as benefiting the community), competency via explicit feedback, and autonomy because students may pick what objects they build and how deep their labor is.

## **2.5.2 Gamifying Cognition: Aligning Incentives with Bloom's Taxonomy**

The Knowledge Marketplace has a tiered currency system that is meant to encourage deeper study. Each level is closely related to Bloom's Taxonomy. At the basic level (Money Points, Levels 1–2), students learn the basics by making clear and correct idea maps and systems. When students reach the application level (Gold Points, Level 3), they must show how these ideas operate in actual engineering projects, such as making a filter for an audio system. They look at numerous ways to solve a problem and talk about the pros and cons of each at the analysis and assessment level (Platinum Points, Levels 4–5). At the highest level (Diamond Points, Level 6), they come up with whole new ways to solve hard and messy technical problems. This tiered framework helps students who want to get the most out of their learning naturally go from memorizing to analyzing and synthesizing.

## **2.6 Roles And Agency: From Consumers to Co-Designers**

### **2.6.1 The Student as Metacognitive Agent and a Source of Value**

Students in VALORIZE have a significantly different role than they do in conventional classes. They don't simply passively take in information; they actively learn and add to the group's knowledge. Creating Knowledge Actively: The major role of students is to make knowledge artifacts, like Primitive and Problem-Solving Maps, that indicate they understand the material and are useful to the course community. This is not simply a section of the test; it's the most important aspect. Assessment occurs not via high-stakes tests, but through the continual evaluation of job quality. Thinking about your own thinking: Students must maintain reflective learning journals that illustrate how they are learning. Using a framework like DAR (Description What Happened; Analysis), students learn to look at their own learning, uncover gaps, and make plans on how to improve. What caused that to happen? What does it mean to reflect? Action Plan What's next? This practice improves metacognitive awareness, which is the ability to think about how your brain works. This is important for being able to adjust. Professional Practice: The framework purposefully incorporates industry-standard protocols into student projects. For example, students may utilize version control systems like Git or GitHub to keep track of their learning artifacts, treating the process of learning as a cycle of development. This serves two purposes: it makes a detailed record of the learning process and teaches students to the professional processes they will use in real life. Portfolio Development: Instead of only being graded on their test scores, students build a portfolio of their greatest work, which they typically share on a personal blog or professional website. This portfolio becomes a real asset that shows prospective employers what you can do and immediately helps you advance in your career.

### **2.6.2 The Educator as Architect, Mentor, and Community Builder**

As Learning Architects, teachers plan the whole learning experience, from planning weekly activities and setting requirements for the Knowledge Marketplace to sequencing tasks to build skills over time and creating social and pedagogical conditions that help students develop their professional identity. This is complicated design job that needs a lot of thought about how people learn. AI takes care of

regular tests (such as giving initial feedback and automated scoring), so teachers have more time and energy to provide good advice. This mentoring is not theoretical; teachers collaborate with students to show them how specialists solve difficult challenges. They ask deep questions to get students to think more deeply, give them feedback that helps them progress, and help each student on their way to become a skilled and moral professional. This design uses two specialized engines that work together to provide a continuous cognitive cycle inside the intelligent engineering paradigm. Figure 1. CKMS-SE System Architecture, which shows the philosophical basis of TISE-VALORIZE, the engines PUDAL and PSKVE that run the system, and the outcomes that were measured. PUDAL engine that puts into action a cycle of constant cognitive progress. This machine is a "micro-evolutionary engine" that pushes constant improvement and change. It is also a PSKVE engine that controls and improves Shared Value Creation (SVC) by making sure that the value provided extends beyond individual learning to the creation of community assets that can be verified. The Core Engine, the AI-based PUDAL Engine for Personalized Learning, and the PSKVE Engine for managing Value Co-Creation (VCC) are the three key parts of this system.

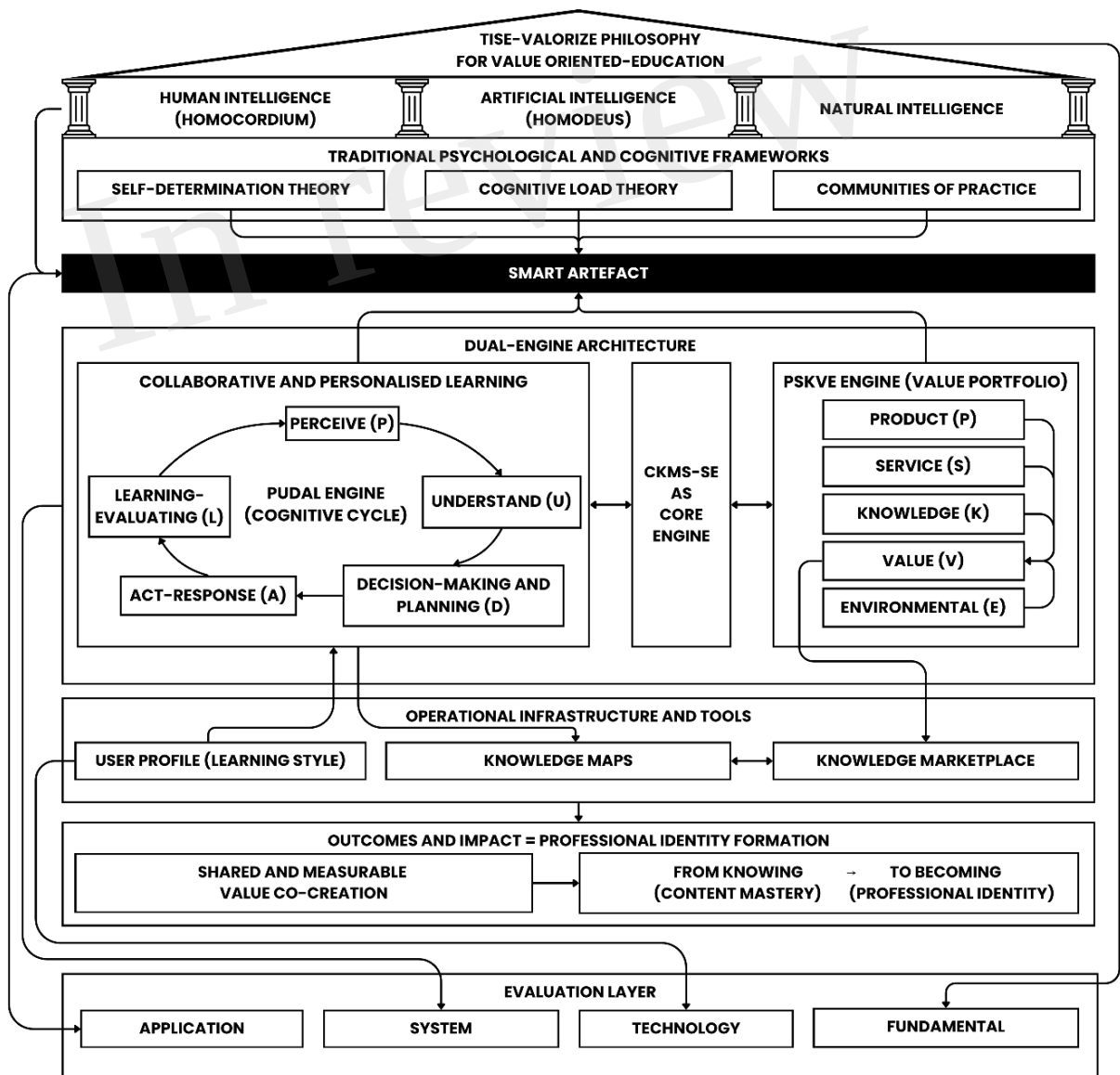


Figure 1. CKMS-SE Architecture

To clarify the empirical scope of the present study, Figure 2 distinguishes between the instructional inputs proposed by the framework, the theoretical mechanisms it draws upon, the performance outcome directly measured in this study, and the broader outcomes proposed for future empirical validation.

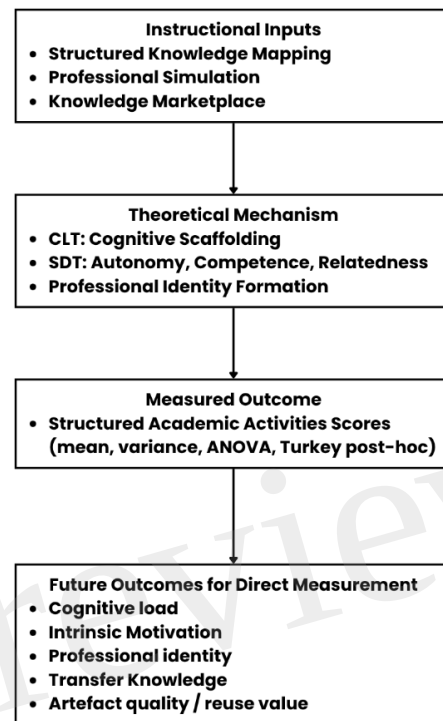


Figure 2. Conceptual scope of the TISE-VALORIZE framework in the present study

### 3 Preliminary Empirical Validation

#### 3.1 Study Design

The empirical component of this paper employed a posttest-only quasi-experimental design with non-equivalent cross-cohort groups. Random assignment was not feasible because the intervention was implemented in naturally occurring course cohorts across different semesters. Accordingly, the findings should be interpreted as preliminary comparative evidence rather than causal proof.

#### 3.2 Participants and Context

The study involved three cohorts of undergraduate engineering students enrolled in the same course context across different semesters. Control Group 1 (C1) consisted of 42 students from the even semester of 2023/2024 and received conventional instruction. Control Group 2 (C2) consisted of 43 students from the odd semester of 2024/2025 and received conventional instruction. The intervention group (E) consisted of 53 students from the odd semester of 2025/2026 and received instruction using the TISE-VALORIZE framework.

Demographic variables such as age, prior achievement, and access to technology were not systematically collected in a form suitable for retrospective controlled analysis. This limits the extent to which cohort differences can be ruled out statistically and is acknowledged as a limitation of the present study.

### 3.3 Assessment and Rubric

Student performance was evaluated through Structured Academic Activities using the same Bloom's Taxonomy-aligned rubric across all cohorts. The rubric emphasized conceptual clarity, mapping linkages, procedural reasoning, and integrated synthesis consistent with the expected quality of engineering problem solving. The assessed artefacts included Primitive Knowledge Maps, Problem-Solving Maps, and integrated synthesis artefacts.

Formal psychometric validation and inter-rater reliability statistics for the rubric were not calculated retrospectively in the present study. Therefore, the rubric should be understood as a common instructional assessment instrument rather than a fully validated research-scale measure. This limitation is acknowledged explicitly in the manuscript.

### 3.4 Statistical Analysis

The analysis included descriptive statistics, one-way ANOVA, Tukey's Honestly Significant Difference (HSD) post-hoc comparisons, Levene's test for homogeneity of variances, and Cohen's *d* for pairwise effect size interpretation. Because no pretest or baseline covariates were collected, ANCOVA could not be applied. Likewise, direct measures of cognitive load, intrinsic motivation, professional identity, and transfer were not available in the present dataset.

Before conducting the omnibus comparison, score distributions in each group were examined using the Shapiro–Wilk test. Because the score distributions deviated from normality, a Kruskal–Wallis test was also conducted as a nonparametric robustness check. One-way ANOVA was retained as the primary omnibus test for comparison with prior reporting, followed by Tukey's HSD for pairwise comparisons and Levene's test for variance differences.

### 3.5 Results

The intervention group showed higher mean performance ( $M = 94.98$ ,  $SD = 7.21$ ) than Control 1 ( $M = 88.70$ ,  $SD = 18.33$ ) and Control 2 ( $M = 86.54$ ,  $SD = 17.05$ ). The intervention group also showed substantially lower variance ( $Var = 52.01$ ) than both control groups (C1:  $Var = 335.78$ ; C2:  $Var = 290.98$ ). These descriptive statistics indicate higher mean scores and lower score dispersion in the intervention group relative to both control groups. This decrease in score dispersion suggests more consistent performance in the intervention group, with fewer students located at the extremes of the score distribution. Table 3 presents the descriptive statistics for structured academic activities performance across the three groups.

Table 3. Descriptive Statistics for Scores on Structured Academic Activities

| Group          | N  | M     | SD    | Var    | Median | IQR          | CV (%) | Min   | Max    |
|----------------|----|-------|-------|--------|--------|--------------|--------|-------|--------|
| Control 1 (C1) | 42 | 88.70 | 18.33 | 335.78 | 96.66  | 86.33-99.25  | 20.7   | 0.00  | 100.00 |
| Control 2 (C2) | 43 | 86.54 | 17.05 | 290.98 | 92.91  | 82.21-96.25  | 19.7   | 1.25  | 98.25  |
| Experiment (E) | 53 | 94.98 | 7.21  | 52.01  | 97.91  | 93.16-100.00 | 7.6    | 65.83 | 100.00 |

Note: Structured Academic Activities scores (0-100 scale); M = mean; SD = standard deviation; Var = variance; IQR = interquartile range; CV = coefficient of variation ( $SD/M \times 100$ ); Min/Max = minimum and maximum observed scores.

The one-way ANOVA indicated a statistically significant difference in Structured Academic Activities scores across the three groups,  $F(2, 135) = 4.41, p = .014, \eta^2 = .061$ . Although statistically significant, the effect size was modest, indicating that group membership was associated with a limited portion of the variance in outcome scores under the conditions of this study. Shapiro–Wilk tests indicated that score distributions in all three groups deviated from normality (Control 1:  $W = 0.636, p < .001$ ; Control 2:  $W = 0.641, p < .001$ ; Experiment:  $W = 0.736, p < .001$ ). Therefore, a Kruskal–Wallis test was conducted as a nonparametric robustness check. The result also indicated a significant group difference,  $H(2) = 18.00, p < .001$ , supporting the conclusion that score distributions differed across groups.

An important result, beyond the higher mean scores, is the substantial reduction in score variability. The intervention group showed a variance of 52.01, compared with 335.78 in Control 1 and 290.98 in Control 2. This pattern indicates lower score dispersion and more consistent performance in the intervention group, whereas the control groups showed substantially greater heterogeneity in outcomes.

Table 4. Summary of ANOVA for Structured Academic Activities scores

Anova: Single Factor

SUMMARY

| Groups         | Count | Sum     | Average     | Variance    |
|----------------|-------|---------|-------------|-------------|
| Control (C1)   | 42    | 3725.4  | 88.7        | 335.7849463 |
| Control (C2)   | 43    | 3721.27 | 86.54116279 | 290.9826153 |
| Experiment (E) | 53    | 5033.87 | 94.97867925 | 52.0050963  |

ANOVA

| Source of Variation | SS          | df  | MS          | F           | P-value    | F crit      |
|---------------------|-------------|-----|-------------|-------------|------------|-------------|
| Between Groups      | 1872.578316 | 2   | 936.2891579 | 4.405265401 | 0.01401709 | 3.063203853 |
| Within Groups       | 28692.71765 | 135 | 212.5386493 |             |            |             |
| Total               | 30565.29597 | 137 |             |             |            |             |

Note: SS = sum of squares; df = degrees of freedom; MS = mean square; F = F-statistic; p = probability value; F\_crit = critical F-value.

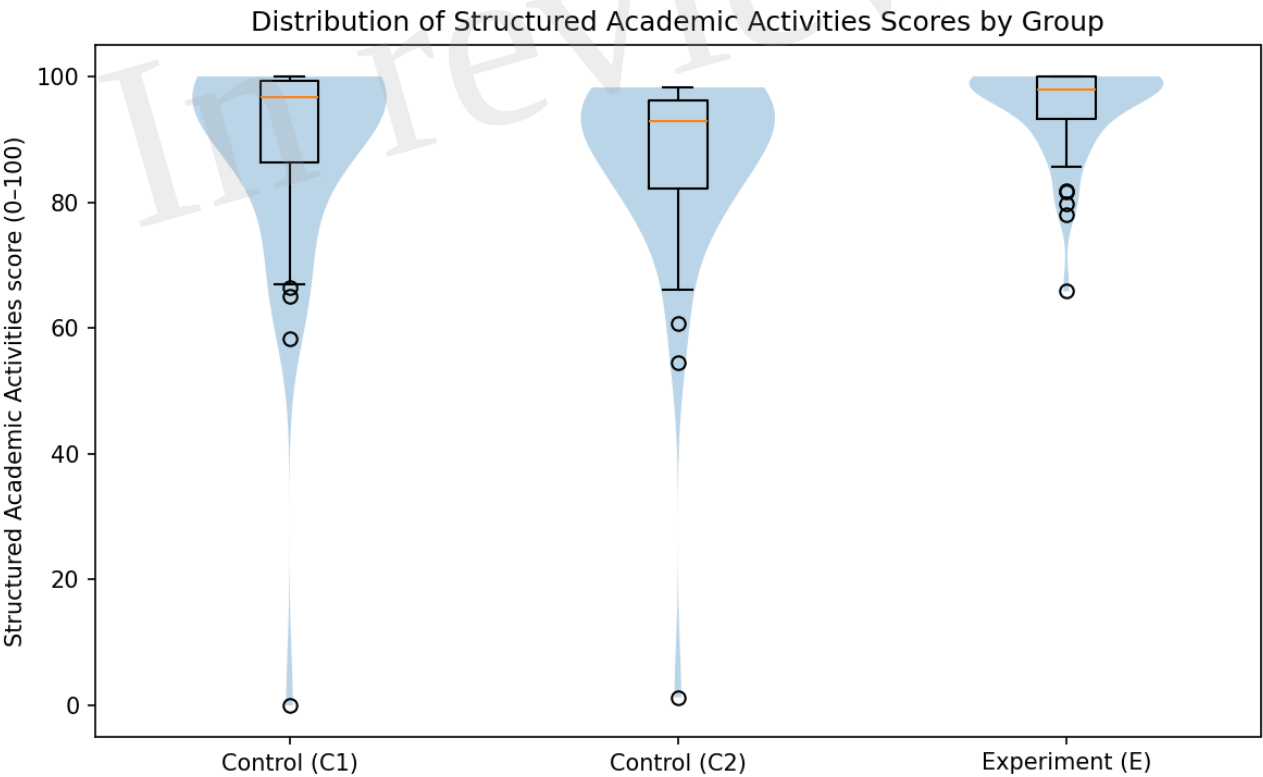
Post-hoc pairwise comparisons using Tukey's Honestly Significant Difference (HSD) test were then conducted to examine which group differences were statistically significant, as shown in Table 5.

Table 5. Comparisons by Pair and Effect Sizes

| Comparison   | Mean Diff | 95% CI         | p-value | d (Cohen's) | Interpretation       |
|--------------|-----------|----------------|---------|-------------|----------------------|
| E versus C1  | 6.28      | [-0.86, 13.42] | .097    | 0.47        | Not significant      |
| E versus C2  | 8.44      | [1.35, 15.53]  | .015    | 0.67        | Significant (medium) |
| C1 versus C2 | -2.16     | [-9.65, 5.34]  | .774    | 0.12        | Not significant      |

404 Note: Mean Diff = difference in group means; 95% CI = 95% confidence interval; p = probability value  
405 (two-tailed); Cohen's d = standardized effect size. Significant at the  $\alpha = .05$  level.

406 Figure 3 presents the score distributions across the three groups using a violin plot with box-plot  
407 overlay. The intervention group shows a more concentrated distribution in the upper score range and a  
408 narrower interquartile spread, whereas the control groups display broader and more dispersed score  
409 distributions. This pattern is consistent with the descriptive statistics and variance analysis, which  
410 suggest more stable performance in the intervention group under the conditions of this study.



411

412

Figure 3. Violin plot with box-plot overlay of Structured Academic Activities scores by group

413 Post-hoc analysis showed a significant pairwise difference between the Experiment and Control 2  
414 groups (Mean Diff = 8.44, 95% CI = [1.35, 15.53],  $p = .015$ , Cohen's  $d = 0.67$ ), indicating a medium  
415 effect size. By contrast, the difference between the Experiment and Control 1 groups did not reach  
416 statistical significance (Mean Diff = 6.28, 95% CI = [-0.86, 13.42],  $p = .097$ , Cohen's  $d = 0.47$ ). There  
417 was also no significant difference between Control 1 and Control 2 (Mean Diff = -2.16, 95% CI = [-



9.65, 5.34],  $p = .774$ , Cohen's  $d = 0.12$ ). Levene's test for homogeneity of variances indicated that score variances differed significantly across groups,  $F(2, 135) = 5.24$ ,  $p = .006$ , supporting the observation that the intervention group showed lower variability than the control groups.

This pattern of results may suggest that the adoption of TISE-VALORIZE, particularly the dual knowledge mapping system (Primitive and Problem-Solving Maps) and the Knowledge Marketplace with tiered payment incentives, produces more uniform and equitable learning outcomes. From the perspective of Cognitive Load Theory (CLT), this pattern may be broadly consistent with the intended scaffolding function of the framework. However, because cognitive load was not directly measured in the present study, no direct claim can be made regarding the framework's effect on cognitive load.

## **4 Discussion: Theoretical Implications and Addressing Counterarguments**

The present study does not directly test all proposed mechanisms of the TISE-VALORIZE framework. Instead, it offers a preliminary empirical indication that the framework may be associated with higher and more consistent performance outcomes, while the proposed motivational, cognitive, and identity-related mechanisms remain to be tested in future studies. Practically, the framework suggests that engineering educators can redesign courses around structured knowledge mapping, professional simulation, and AI-supported feedback to improve consistency of student outcomes.

### **4.1 Interpretation of Findings**

The present study provides preliminary empirical evidence that the TISE-VALORIZE framework may be associated with higher and more consistent performance outcomes in a digital engineering education context. The intervention group showed both higher mean scores and lower score dispersion than the two conventional instruction groups. However, the observed effect size was modest, which suggests that the framework may have contributed to performance differences under the conditions of this study, but it does not account for most of the observed variance.

These findings are broadly consistent with the instructional logic of the framework. In particular, the use of structured knowledge mapping, professional simulation, and contribution-oriented learning activities may have supported more stable performance across students. At the same time, the present study does not directly test the proposed motivational, cognitive-load, or identity-related mechanisms. Therefore, these findings should be interpreted as preliminary performance-based evidence rather than direct confirmation of the full theoretical model.

### **4.2 Comparison with Prior Work**

The present findings are broadly aligned with prior research suggesting that structured scaffolding and meaningful participation can support learning in higher education. From a Cognitive Load Theory perspective, the framework is consistent with the use of structured representations to support schema development. From a Self-Determination Theory perspective, the framework is also consistent with the idea that meaningful contribution, perceived choice, and visible progress may support engagement. However, because these mechanisms were not directly measured here, the present study should be understood as conceptually aligned with prior theory rather than as a direct empirical test of those theories.

### 4.3 Limitations

This study has several important limitations. First, the design was a posttest-only quasi-experimental comparison with non-equivalent cross-cohort groups rather than a randomized controlled evaluation. As a result, cohort effects and semester-level confounding cannot be ruled out. Second, no pretest or baseline covariates were available, which limits causal interpretation and prevented the use of ANCOVA. Third, the empirical analysis relied on performance outcomes from Structured Academic Activities and did not directly measure cognitive load, intrinsic motivation, professional identity, or transfer. Fourth, demographic variables suitable for controlled comparison were not systematically available. Fifth, Structured Academic Activities were assessed by a single course assessor using the same Bloom's Taxonomy-aligned rubric across cohorts. Formal inter-rater reliability was therefore not applicable. However, no retrospective intra-rater reliability check was conducted, and formal psychometric validation evidence for the rubric was not calculated retrospectively. This limits the extent to which score comparisons can be attributed solely to the intervention. These limitations mean that the findings should be interpreted as preliminary and performance focused.

### 4.4 Future Directions

Future studies should test the framework using stronger designs, including matched or randomized groups, pretest-posttest comparisons, and larger multi-institutional samples. In addition, future work should include validated instruments for cognitive load, intrinsic motivation, professional identity, and transfer, so that the proposed mechanisms of the framework can be examined directly. Secondary indicators such as artefact quality, marketplace participation, and reuse value may also provide a richer picture of how value-oriented learning operates in practice.

## 5 Conclusion

This study proposes the TISE-VALORIZE framework as a value-oriented approach to digital engineering education that integrates structured knowledge mapping, professional simulation, and contribution-based learning. In this preliminary posttest-only quasi-experimental comparison, the intervention group showed higher mean performance and lower score dispersion than two conventional instruction groups, although the observed effect size was modest. Practically, the framework suggests a possible direction for redesigning engineering courses around structured representation, authentic participation, and AI-supported feedback. Future research should test the framework with stronger designs and direct measures of cognitive load, motivation, professional identity, and transfer.

## 6 Additional Requirements

The datasets generated for this study are available on request to the corresponding author.

## 7 Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

## 8 Author Contributions

All authors approved the final manuscript and agreed to be accountable for all aspects of the work.

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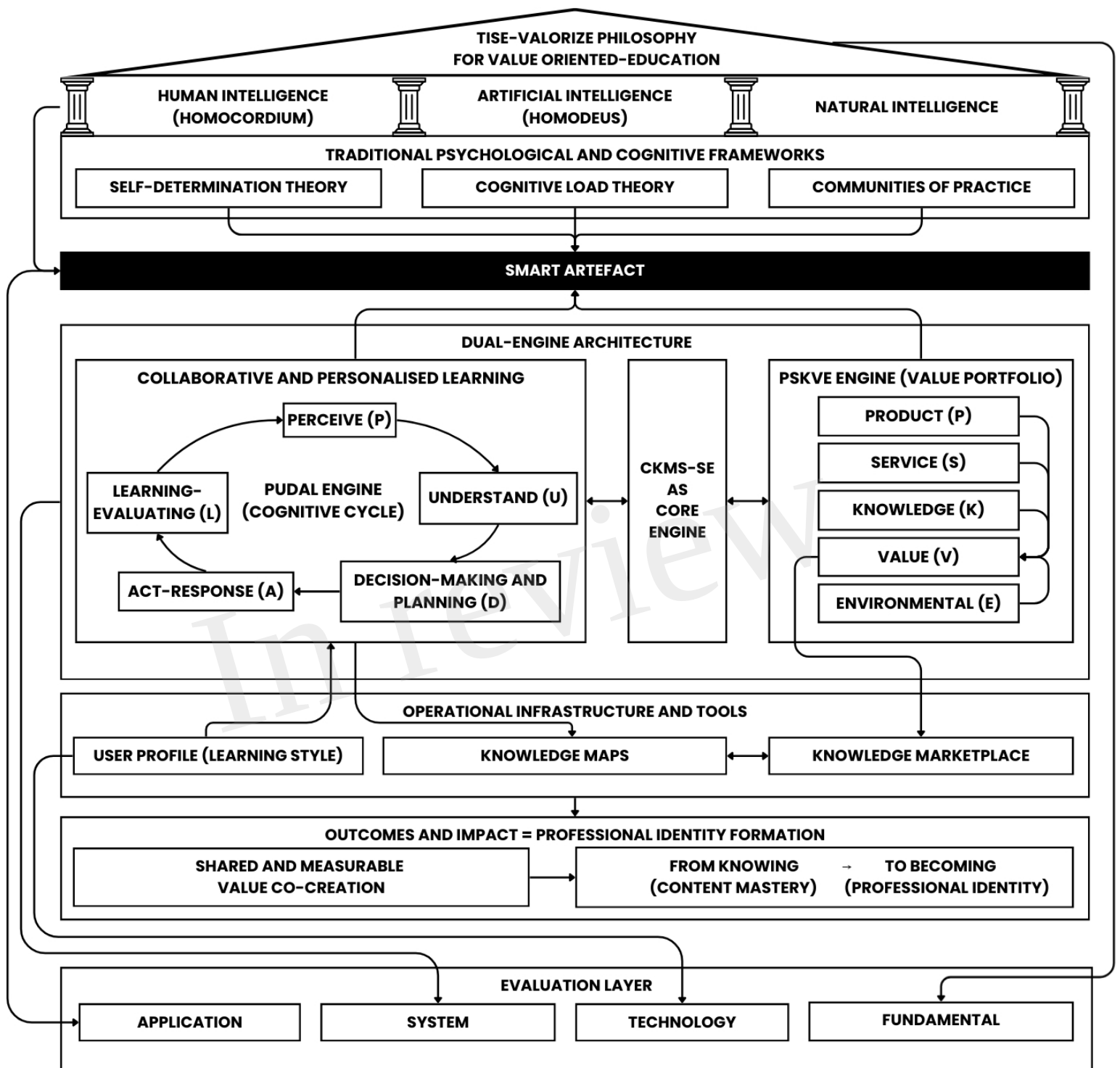


Figure 2.JPEG

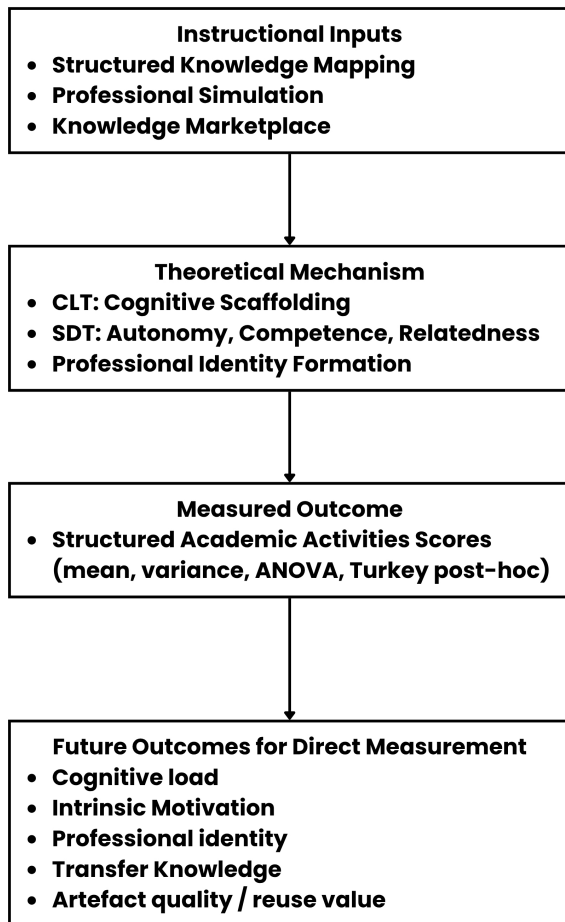
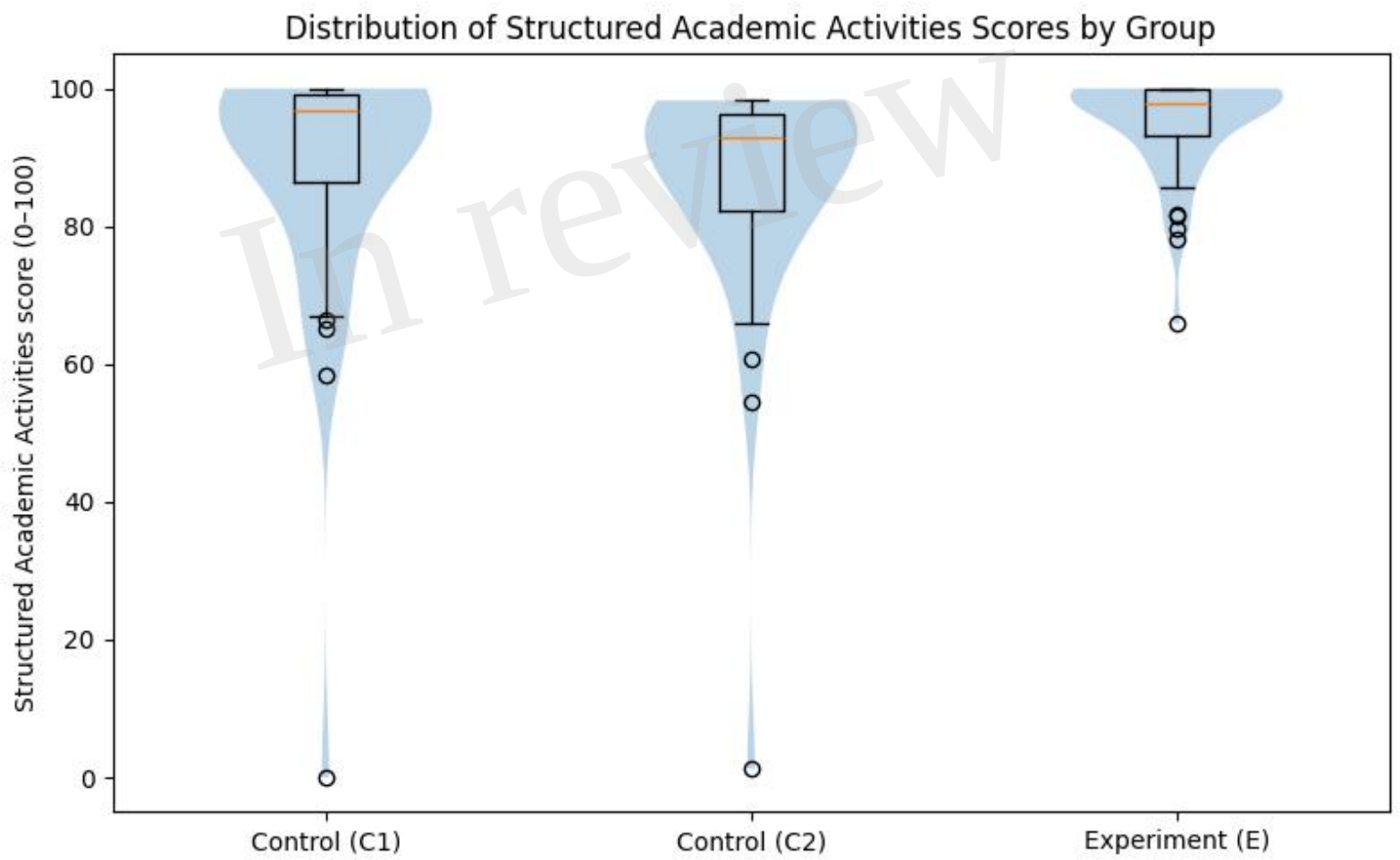


Figure 3.JPEG





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Meliana Christianti Johan \*, Armein Z. R. Langi , I Gusti Bagus Baskara Nugraha , Radiant Victor Imbar and Partindungan Sinaga

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